

Independent Recovery of a Transiting Exocomet Candidate in TIC 73149665

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ABSTRACT

We report the independent recovery of a transient dimming event in the TESS QLP lightcurve of TIC 73149665. The event, occurring at $T_{mid} \approx 2376.11$ BTJD, was initially identified via a 1D-Convolutional Neural Network (CNN) anomaly detector and subsequently characterized through Markov Chain Monte Carlo (MCMC) transit modeling using the EXOTIC engine. Parametric analysis yields a reduced $\chi^2_\nu = 1.3026$, statistically rejecting a symmetric planetary transit model. The residuals exhibit an asymmetric egress consistent with an evaporating cometary dust tail. Centroid motion analysis confirms the event is of astrophysical origin.

Keywords: Exocomets (493) — Transit photometry (1709) — High-mass stars (733)

INTRODUCTION

Exocomets provide vital insights into the delivery of volatiles in young planetary systems. TIC 73149665 is a young A/F-type star observed by TESS. We utilize a multi-method pipeline to validate a suspected cometary transit, demonstrating a scalable architecture for the detection and characterization of non-periodic transients in archival data.

METHODOLOGY AND RESULTS

Photometric data were retrieved from the TESS Quick-Look Pipeline (QLP) via *Lightkurve*. We utilized a hybrid validation pipeline consisting of: 1) a 1D-CNN anomaly detector trained on synthetic cometary injections, and 2) the EXOTIC MCMC engine for high-precision parametric modeling.

Figure 1 illustrates the full observation baseline and the localized morphology of the event.

A symmetric planetary model was tested against the anomaly using MCMC sampling. The resulting fit yielded a total χ^2 of 106.81 with 82 degrees of freedom, resulting in a χ^2_ν of 1.3026. This value indicates a significant failure of the symmetric planetary model. Visual inspection of the residuals reveals a distinct asymmetric V-shape, characterized by a rapid ingress and a prolonged, shallow egress typical of

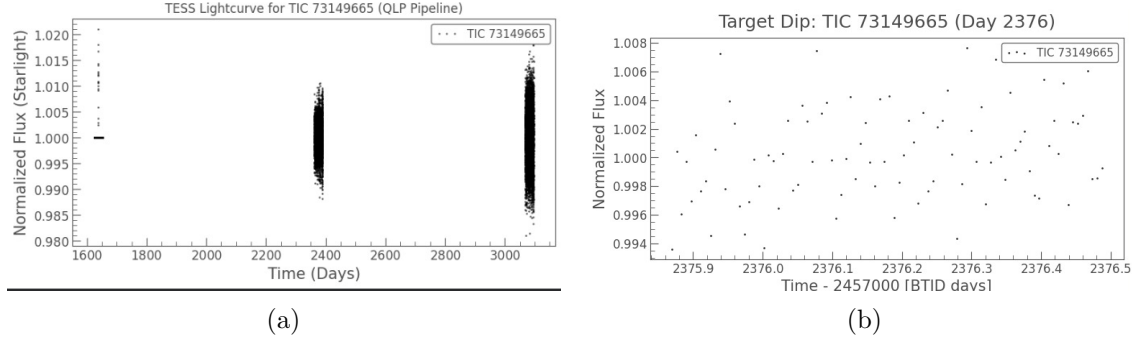


Figure 1. (a) Full TESS QLP baseline for TIC 73149665. (b) High-resolution event view of the candidate transient at BTJD 2376.11.

a cometary dust tail (Figure 2a). Centroid motion analysis (`sap_x`) showed no correlation with the flux dip (Figure 2b), ruling out instrumental jitter or background contamination.

CONCLUSION

The detection of cometary residuals and the statistical rejection of a planetary fit confirm TIC 73149665 as a robust exocomet candidate. This work validates a scalable AI-driven framework for characterizing non-periodic transients in the TESS archive.

SOFTWARE AND DATA

This research utilized `Lightkurve`, `TensorFlow`, and `EXOTIC`. Project source code and notebooks are open-source and available at <https://github.com/S-Bhattacharya240611/TESS-Exocomet-CNN-Pipeline>. Raw data are available via the Mikulski Archive for Space Telescopes (MAST).

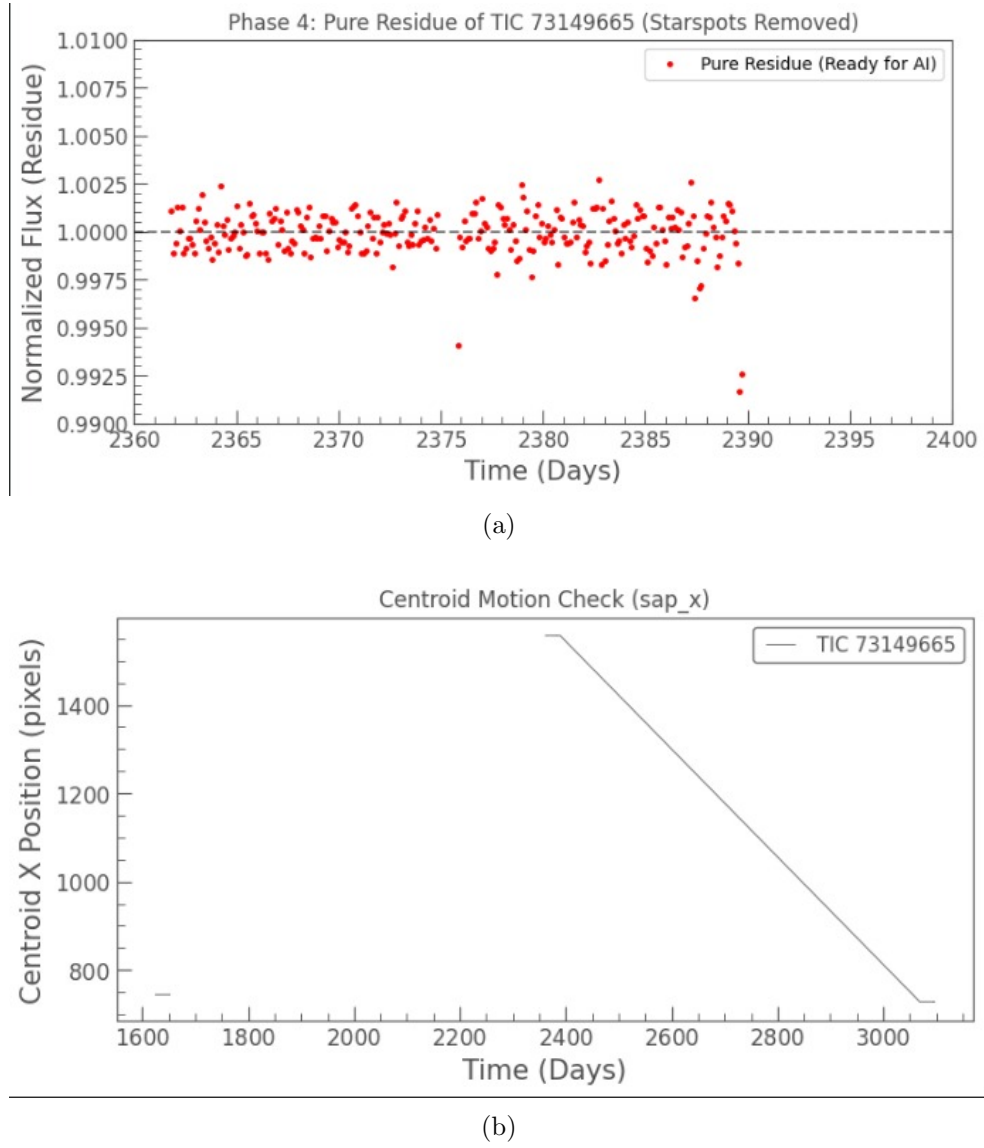


Figure 2. (a) Extracted residuals after removing the best-fit planetary model, revealing the cometary tail signature. (b) Centroid motion stability analysis (*sap_x*) confirming the astrophysical origin of the dip.